Top 40 Most Repeated & High-Probability Exam Questions

GTU Diploma Sem-1 Hub · Bold letters (**a**, **i**, **j**, **k**, etc.) denote vectors.

Unit 1: Determinants & Matrices

1. Solve the determinant equation for x:

- $$\begin{vmatrix} x & 4 & 4 \\ 4 & x & 4 \\ 4 & 4 & 0 \end{vmatrix} = 0$$

2. Solve the determinant equation for x:

- $$\begin{vmatrix} 4 & 2 & x \\ 1 & 5 & -1 \\ 3 & 2 & 3 \end{vmatrix} = 4$$

3. Find the value of the 3×3 determinant:

- $$\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix}$$

4. Solve the determinant equation for x:

- $$\begin{vmatrix} x & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 3 \end{vmatrix} = 3$$

Unit 2: Logarithms & Functions

5. Prove that:

- $$\frac{1}{\log_{24} 12} + \frac{1}{\log_{8} 12} + \frac{1}{\log_{9} 12} = 3$$

6. Prove that:

- $$\frac{1}{\log_{12} 120} + \frac{1}{\log_{2} 120} + \frac{1}{\log_{5} 120} = 1$$

7. Prove that:

- $$\log_{b4}(a^3) \cdot \log_{c4}(b^3) \cdot \log_{a4}(c^3) = \frac{27}{64}$$

(bases are 4th powers, arguments are cubes)

8. Prove that:

- $$3 \log\left(\frac{3}{4}\right) + ½ \log\left(\frac{16}{9}\right) - \log\left(\frac{1}{48}\right) - 2 \log(\ldots)$$

Right-most term of Q8 was cut off in the photo — recheck your notes for the final log term.

9. Solve for x:

- $$\frac{\log x \times \log 16}{\log 256} = \log 32$$

10. If log(x − y) = log 2 + ½ log x + ½ log y, prove that:

- $x^2 + y^2 = 6xy$
- $$\frac{x}{y} + \frac{y}{x} = 6$$

11. If the following are equal, prove that xyz = 1 and $x^a \cdot y^b \cdot z^c = 1$:

- $$\frac{\log_{10} x}{b - c} = \frac{\log_{10} y}{c - a} = \frac{\log_{10} z}{a - b}$$
- $xyz = 1$
- $x^a \cdot y^b \cdot z^c = 1$

12. Prove that:

- $$\log\left(\frac{x^p}{x^q}\right) + \log\left(\frac{x^q}{x^r}\right) + \log\left(\frac{x^r}{x^p}\right) = 0$$

13. If f(x) = (1−x)/(1+x), prove that:

- $$f(x) + f\left(\frac{1}{x}\right) = 0$$
- $$f(x) - f\left(\frac{1}{x}\right) = 2f(x)$$

14. If $f(x) = e^x$, prove that:

- $f(x+y) = f(x) \cdot f(y)$
- $f(x-y) = f(x)/f(y)$

Unit 3: Trigonometry

15. Prove that:

- $\tan 35° + \tan 10° + \tan 35° \cdot \tan 10° = 1$

16. Prove that:

- $$\tan(57°) = \frac{\cos 12° + \sin 12°}{\cos 12° - \sin 12°}$$

17. Prove that:

- $$\tan(70°) = \frac{\cos 25° + \sin 25°}{\cos 25° - \sin 25°}$$

18. Prove that:

- $8 \cos(20°) \cos(40°) \cos(80°) = 1$

19. Prove that:

- $$\frac{\sin(\pi/2+\theta)}{\cos(2\pi-\theta)} + \frac{\tan(\pi+\theta)}{\cot(\pi/2-\theta)} + \frac{\sec(3\pi/2+\theta)}{\mathrm{cosec}(\pi-\theta)} = \ldots$$

Right-hand side of Q19 was cut off in the photo — verify against your notes.

20. Prove that:

- $$\frac{\sin(\pi-\theta)\cdot\cot(\pi/2+\theta)\cdot\sin(3\pi/2-\theta)}{\cos(3\pi/2+\theta)\cdot\cos(\pi/2+\theta)\cdot\tan(2\pi-\theta)} = \ldots$$

Right-hand side of Q20 was cut off in the photo — verify against your notes.

21. Prove that:

- $$\frac{\sin(180°-x)}{\cos(90°+x)} + \frac{\mathrm{cosec}(180°-x)}{\sec(90°+x)} + \frac{\tan(180°-x)}{\cot(90°+x)} = \ldots$$

Right-hand side of Q21 and the last term were partly cut off in the photo — verify against your notes.

22. Prove that:

- $$\frac{\cos\theta + \cos 3\theta + \cos 5\theta}{\sin\theta + \sin 3\theta + \sin 5\theta} = \cot 3\theta$$

23. Prove that:

- $$\frac{\sin\theta + \sin 2\theta + \sin 3\theta}{\cos\theta + \cos 2\theta + \cos 3\theta} = \tan 2\theta$$

24. Prove that:

- $$\tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{3}\right) = \pi/4$$

25. Evaluate:

- $\tan(75°)$
- $$4\sin^2(\pi/3) - \frac{3}{4}\tan^2(\pi/6) + \frac{4}{3}\cot^2(\pi/6)$$

26. Draw the trigonometric graph for either:

- $y = \cos x, \quad -\pi/2 \le x \le \pi/2$
- $y = \sin x, \quad 0 \le x \le \pi$

Unit 4: Vectors

27. If $\mathbf{a} = 2\mathbf{i} - \mathbf{j}$ and $\mathbf{b} = \mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$, find $|(\mathbf{a}+\mathbf{b})\times(\mathbf{a}-\mathbf{b})|$.

28. If $\mathbf{x} = 3\mathbf{i} - 8\mathbf{j}$ and $\mathbf{y} = 5\mathbf{j} - 4\mathbf{k}$, find $|(\mathbf{x}+\mathbf{y})\times(\mathbf{x}-\mathbf{y})|$.

29. Find the linear combination magnitude:

- If $\mathbf{a}=\mathbf{i}+2\mathbf{j}+\mathbf{k}$, $\mathbf{b}=2\mathbf{i}-3\mathbf{j}+\mathbf{k}$, $\mathbf{c}=-2\mathbf{i}-\mathbf{j}+5\mathbf{k}$, find $|2\mathbf{a}+3\mathbf{b}-\mathbf{c}|$.
- If $\mathbf{a}=\mathbf{i}-2\mathbf{j}+4\mathbf{k}$, $\mathbf{b}=-3\mathbf{i}+\mathbf{j}-4\mathbf{k}$, $\mathbf{c}=\mathbf{i}+2\mathbf{j}-4\mathbf{k}$, find $|5\mathbf{a}+3\mathbf{b}-2\mathbf{c}|$.

30. Find the scalar value P or x for which the vectors are perpendicular:

- $2\mathbf{i}+3\mathbf{j}-5\mathbf{k}$ and $P\mathbf{i}+\mathbf{j}+3\mathbf{k}$ are perpendicular.
- $2\mathbf{i}-3\mathbf{j}+5\mathbf{k}$ and $x\mathbf{i}-6\mathbf{j}-8\mathbf{k}$ are perpendicular.

31. Show that the angle between the given vectors satisfies the specified value:

- Between (1,1,−1) and (2,−2,1) is $\sin^{-1}(\sqrt{26/27})$.
- Between **i**−**j** and 2**i**+**j**−**k** is $\cos^{-1}(1/(2\sqrt{3}))$.
- Between **a**=**i**+2**j** and **b**=**i**+**j**+3**k** is $\sin^{-1}(\sqrt{46}/(5\sqrt{5}))$.

32. Find the unit vector perpendicular to both vectors **x**=(3,1,2) and **y**=(2,−2,4).

33. Find the total work done by constant forces:

- Under forces (1,2,1) and (2,−1,0), a particle moves from (−1,2,1) to (2,3,−1).
- Under forces **i**+3**k**, 2**i**+**k**, and 2**i**+**j**−3**k**, a particle moves from (1,2,−3) to (5,3,7).

Unit 5: Coordinate Geometry (Line & Circle)

34. Find the equation of the line passing through:

- Passing through (−6,2) and parallel to the line joining (−3,1) and (−5,2).
- Passing through (−3,2) and parallel to the line $x-2y+1=0$.
- Passing through the points (1,2) and (2,1).

35. Determine the value of m if the lines $5x-my=3$ and $2x+3y=4$ are:

- Parallel
- Perpendicular

36. Find the angle between the lines $x+y+1=0$ and $2x+3y+7=0$.

37. Find the equation of the circle under the following conditions:

- Center at (3,−1) and passing through (−1,2).
- Center at (2,3) and circumference 4π.
- Center at (2,−3) and radius 3.

38. Find the equation of the tangent and normal to the circle:

- To $x^2+y^2-4x+2y+3=0$ at point $(-1,2)$.
- To $x^2+y^2-2x-9=0$ at point $(2,3)$.

Unit 6: Limits (Calculus Fundamentals)

39. Evaluate the algebraic & rational limits:

- $\lim_{x\to2} \dfrac{x^3-3x^2+5x-6}{x^3-8}$
- $\lim_{x\to0} \dfrac{\sqrt{(25+x)} - 5}{x}$
- $\lim_{x\to0} \dfrac{\sqrt{(x+4)} - 2}{x}$
- $\lim_{x\to-3} \dfrac{x^3+27}{x^2+5x+6}$
- $\lim_{x\to3} \dfrac{x^3-27}{\sqrt{(x^3)} - \sqrt{(3^3)}}$
- $\lim_{n\to\infty} (\sqrt{(n^2+n+1)} - n)$

40. Evaluate the standard trigonometric & exponential limits:

- $\lim_{x\to0} \dfrac{1 - \cos x}{x^2}$
- $\lim_{x\to0} \dfrac{4^x - 3^x}{x}$
- $\lim_{x\to0} \dfrac{2^x - 5^x}{x}$
- $\lim_{x\to\infty} (1 + 5/x)^x$